

Standard errors with very small or very large datasets

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Problems with standard inference

Degrees of freedom correction

Alternative alternatives

- Consider regression of Y_i onto $X_i = (D_i, W_i)$, $k = \dim(X_i)$. **When are Eicker-Huber-White (EHW) and Liang-Zeger (LZ) standard errors unreliable?**
- Suppose $E[Y_i | X_i = x] = d\beta + w'\gamma$, and that we want to do inference on β conditional on X
 - Perhaps not the best approach for ensuring causal or descriptive interpretation for β robust to non-linearity of regression function.
 - But it makes it easy to think through hiccups with standard inference.

Regularity conditions for central limit theorem (CLT)

No fat tails $E[\epsilon_i^{2+\eta} | X]$ is bounded for some $\eta > 0$.

Low partial leverage $\max_i H_{\tilde{D},ii} \rightarrow 0$ (i.e. no outliers in X_i).

Regularity conditions for inference

For consistency of $\hat{V}_{\text{EHW}}$ need to strengthen the leverage condition. Sufficient conditions:

Leverage for inference Either $\max_i H_{X,ii} \rightarrow 0$ or $k \max_i H_{\tilde{D},ii} \rightarrow 0$.

- First version ensures consistency of full regression function. Violated with fixed effects. Conditions allow for “high-dimensional” setting with $k \rightarrow \infty$, but trouble if $k \asymp n$.

Lemma

Suppose that the above conditions hold. Then EHW standard errors lead to asymptotically valid inference.

What can go wrong?

1. CLT fails

- Look at outliers. Can winsorize, but changes interpretation of β .
 - Alternative inference procedures exist (Müller 2023).
- Look at **partial** leverage $\max_i H_{\tilde{D},ii}$

2. EHW variance estimator is not consistent: it displays finite-sample bias or substantial sampling variability: t -stats not normal, leading to undercoverage.

- Natural solution is to bias-correct estimator and use degrees of freedom (DoF) correction

Example 1

Suppose that $D_1 = C\sqrt{n}$ for some constant C , while $D_i = 1$ if $i > 1$, and that $W_i = 1$. Then $H_{X,11} = 1$, while $H_{X,ii} = 1/(n-1)$ for $i > 1$. $\hat{\beta}$ is $\sqrt{n}$ -consistent, but not asymptotically normal unless ϵ_1 happens to be normal.

Bo Honoré's outlier detection method

Wish to check whether first observation outlier, so set $D_i = \mathbb{1}\{i = 1\}$. W_i are well-behaved controls. Then (i) $H_{X,11} = 1$ (ii) $\hat{\epsilon}_1 = 0$ (iii) $\hat{\gamma}$ consistent, and $\hat{\beta}$ converges to $\beta + \epsilon_1$, and (iv) t -statistic for $\hat{\beta}$ based on EHW standard errors converges to $\pm\infty$ irrespective of value of β .

Berhens-Fisher problem (Behrens 1929; Fisher 1939)

n_1 observations are treated, n_0 controls. Only covariates are intercept. Assume $\epsilon_i \mid D_i \sim \mathcal{N}(0, \sigma^2(D_i))$. Clear that even if n large, effective sample size small if $\min\{n_1, n_0\}$ is small. Leverage reflects this: $H_{X,ii} = 1/n_{D_i}$.

- More complicated version of this problem arises in differences-in-differences contexts with a few treated observations.

Clustering introduces additional complications:

- Sample size is determined by number of clusters S : asymptotics are as $S \rightarrow \infty$.
- Rate of convergence depends heterogeneity in cluster sizes and on within-cluster correlation structure. It'll be at most $n^{-1/2}$, but it can be even much slower than $S^{-1/2}$.
- Necessary that $\max_i H_{\tilde{D},ii} \rightarrow 0$ for CLT to hold. But what matters is leverage of whole cluster, so sufficient leverage condition substantially stronger.

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In general:

$$B = E[\hat{V}_{\text{EHW},11} | X] - \mathcal{V}_{\text{cx},11} = \frac{\sum_i E[\hat{\epsilon}_i^2 - \sigma^2(X_i) | X_i] \ddot{D}_i^2}{(\sum_i \ddot{D}_i^2)^2} \asymp \frac{\sum_i H_{X,ii} \ddot{D}_i^2}{(\sum_i \ddot{D}_i^2)^2}$$

which is where the leverage for inference condition comes from. Under homoskedastic errors:

$$E[\hat{V}_{\text{EHW}} | X] - \mathcal{V}_{dc} = \sigma^2 n (X'X)^{-1} \sum_i (H_{X,ii} - 1) X_i X_i' (X'X)^{-1} \leq 0.$$

Simple solution is to replace EHW with (MacKinnon and White 1985)

$$\hat{V}_{HC2} = n (X'X)^{-1} \sum_i \frac{\hat{\epsilon}_i^2}{1 - H_{X,ii}} X_i X_i' (X'X)^{-1},$$

- Using HC2 estimator solves bias issue, but another issue is variance: reason for using t -distribution critical values under homoskedastic normal errors.
- Makes sense to also use DoF correction with heteroskedastic errors.
- Simplest approach is to use ν DoF, where ν matches first 2 moments of variance estimator under homoskedasticity (Satterthwaite 1946)
 - Formula in lecture notes refinement that provides finite-sample guarantees instead of heuristics
- **Key point:** DoF adjustment reflects distribution of covariates and hence any leverage issues

- Similar adjustments can be applied under clustering.
- Bell and McCaffrey (2002) generalize both bias and DoF correction, based on matching DoF under homoskedastic Gaussian benchmark
- But can use other working models. See Imbens and Kolesár (2016) for details, and Hansen (2025) for refinement.
- These adjustments are heuristic, but tend to work well in practice.
 - Look out for a forthcoming working paper by Kolesár and Müller that replaces heuristics with a DoF correction that has finite-sample guarantees.

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Alternative alternatives

- Possible to construct variance estimators that are exactly unbiased.
- Approach 1: use Hadamard products (Dobriban and Su 2024; Cattaneo, Jansson, and Newey 2018). Involves inverting $n \times n$ matrices.
- Leave-out approach (Kline, Saggio, and Sølvesten 2020; Jochmans 2022): estimate $\sigma^2(X_i)$ in variance formula not by $(Y_i - X_i' \hat{\theta})^2$ used by EHW, but by unbiased estimator

$$\check{\sigma}_i^2 = Y_i(Y_i - X_i' \hat{\theta}_{-i}) = \frac{Y_i(Y_i - X_i' \hat{\theta})}{1 - H_{X,ii}}$$

Downside: can be noisy

- Formally, both approaches with when $p \asymp n$.

- Popularized by Cameron, Gelbach, and Miller (2008).
- Confidence intervals have to be computed by test inversion:
 1. To test the null $\ell' \theta = c$, compute the OLS estimate θ subject to this restriction, obtaining the restricted estimate $\hat{\theta}_r$ and residuals $\hat{\epsilon}_i^r$.
 2. Let $Y_i^* = X_i' \hat{\theta}_r + g_i^* \hat{\epsilon}_i^r$, where $g_i^* \in \{-1, 1\}$ (with equal probability), and let $X_i^* = X_i$. Compute $\hat{\theta}^*$ using ordinary least squares (OLS) in this bootstrap sample.
 3. As a critical value for the test statistic $|\ell' \hat{\theta} - c|$, use the $1 - \alpha$ quantile of $|\ell' (\hat{\theta}^* - \hat{\theta}_r)|$
- Canay, Santos, and Shaikh (2021) show formally that this method works even with a fixed number of clusters, but need strong homogeneity conditions on distro on covariates across clusters

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